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RESEARCH ARTICLE

Phylogenetic variations in a novel family of hyperstable apple snail egg proteins: insights into structural stability and functional trends

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ABSTRACT

The relationship between protein stability and functional evolution is little explored in proteins purified from natural sources. Here, we investigated a novel family of egg proteins (Perivitellin-1, PV1) from *Pomacea* snails. Their remarkable stability and clade-related functions in most derived clades (Canaliculata and Bridgesii) make them excellent candidates for exploring this issue. To that aim, we studied PV1 (PpaPV1) from the most basal lineage, Flagellata. PpaPV1 displays unparalleled structural and kinetic stability, surpassing PV1s from derived clades, ranking among the most hyperstable proteins documented in nature. Its spectral features contribute to a pale egg coloration, exhibiting a milder glycan binding lectin activity with a narrower specificity than PV1s from the closely related Bridgesii clade. These findings provide evidence for substantial structural and functional changes throughout the genus' PV1 evolution. We observed that structural and kinetic stability decreased in a clade-related fashion and was associated with large variations in defensive traits. For instance, pale PpaPV1 lectin turns potent in the Bridgesii clade, adversely affecting gut morphology, while giving rise to brightly colored PV1s providing eggs with a conspicuous, probably warning signal in the Canaliculata clade. This work provides a comprehensive comparative analysis of PV1s from various apple snail species within a phylogenetic framework, offering insights into the interplay among their structural features, stability profiles and functional roles. More broadly, our work provides one of the first examples from natural evolution showing the crucial link among protein structure, stability and evolution of new functions.

KEY WORDS: Protein evolution, Egg protection, Defensive proteins, Reproductive strategy, Hyperstable proteins

INTRODUCTION

Protein stability affects functional aspects and evolvability, establishing a tight balance between increased functionality through the accumulation of mutations and the ability to maintain an adequate level of stability (Gershenson et al., 2014). However, there are few examples from nature illustrating these tradeoffs, as opposed to directed evolution in the laboratory (Zheng et al., 2020) or theoretical models (Agozzino and Dill, 2018; Tokuriki and Tawfik, 2009a; Zeldovich et al., 2007).

Apple snails (*Pomacea* spp.) are an emerging model for evolutionary studies because of their high diversity, ancient history and wide geographical distribution (Hayes et al., 2009; Sun et al., 2019). These are amphibious snails that have evolved an unusual reproductive strategy, laying eggs outside the water (Hayes et al., 2009). This transition to terrestrial egg laying went along with the acquisition of remarkable molecular and biochemical changes, particularly their reproductive egg proteins (Ip et al., 2019). *Pomacea* belongs to the family Ampullariidae, that includes aquatic and amphibious apple snails (Fig. 1A). Basal genera are entirely aquatic and deposit gelatinous, non-pigmented or poorly pigmented egg masses (Hayes et al., 2009). In contrast, *Pomacea*, the most derived genus, lays conspicuous, carotenoid-pigmented egg masses. Its adaptation of terrestrial egg deposition is considered a crucial event for avoiding aquatic predation and/or parasitism, contributing to *Pomacea* being the most species-rich and widely distributed genus in Ampullariidae (Fig. 1A) (Hayes et al., 2009).

Pomacea species are distributed in four well-supported clades: Flagellata, Effusa, Bridgesii and Canaliculata (Hayes et al., 2009) (Fig. 1B). The Canaliculata clade includes the most studied apple snails, such as *Pomacea maculata* Perry 1810, and *Pomacea canaliculata* (Lamarck 1822). These two invasive species are pests of aquatic crops and intermediate hosts for human parasitic diseases (Hayes et al., 2015). The Canaliculata clade is the most derived within *Pomacea*. Its sister clade, Bridgesii, includes non-invasive species such as *Pomacea scalaris* (d'Orbigny 1835) and *Pomacea diffusa* Blume 1957 (Hayes et al., 2008). The most basal clade, Flagellata, has non-invasive species including *Pomacea patula catemacensis* (Baker 1922), which is restricted in its distribution to Catemaco Lake in Mexico (Diupotex-Chong et al., 2004). Multiple lines of evidence on adaptive evolution in apple snail egg proteins contribute to our understanding of how the aquatic gastropod ancestors successfully invaded terrestrial habitats (Ip et al., 2019). The traditional view of proteins possessing absolute functional specificity and a single fixed structure conflicts with their marked ability to adapt and evolve new functions and structures (Tokuriki and Tawfik, 2009b).

In some animal genera, orthologous proteins play similar roles but undergo major functional adaptations. A family of reproductive egg carotenoproteins from apple snails called Perivitellin-1 (PV1) showcases this phenomenon. These proteins have no sequence similarity with proteins of organisms outside the ampullariid family, suggesting that PV1s have arisen by duplication of ancient genes (Fig. 1A) (Sun et al., 2012, 2019). PV1 orthologs have only been isolated and studied in the most derived Canaliculata and Bridgesii clades of *Pomacea* genus, while no information is available on the basal ones; Fig. 1B summarizes the clades, species and PV1 proteins analyzed in this study. PV1s share several structural features: they provide coloration to eggs by associated carotenoid

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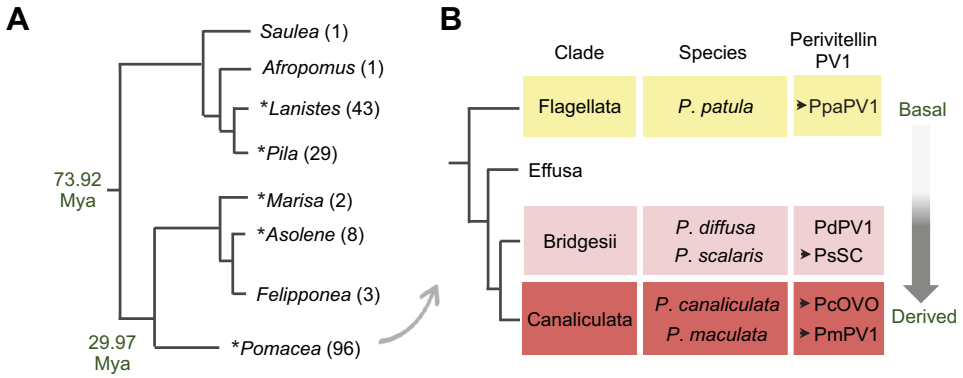


Fig. 1. Simplified phylogenetic hypothesis for the Ampullariidae and Pomacea. (A) Phylogenetic tree of Ampullariidae. The number of currently valid species is shown in parentheses. (B) Phylogenetic tree of the *Pomacea* genus, highlighting the clades, species and Perivitellin-1 (PV1) proteins characterized. The *Pomacea* Flagellata clade, the most basal (shaded yellow) includes *Pomacea patula catemacensis* whose PV1 we dubbed PpaPV1. The Bridgesii clade (shaded pink) includes *Pomacea scalaris* (whose PV1 is called PsSC) and *Pomacea diffusa* (whose PV1 is called PdPV1); the most derived clade is Canaliculata (shaded red), represented by *Pomacea maculata* (whose PV1 is PmPV1) and *Pomacea canaliculata* (whose PV1 is PcOVO). Arrowheads indicate the PV1s analyzed in this study. *Pomacea* phylogeny is based on Hayes et al. (2015). *Species/genus containing reported PV1 homologous sequences (Ip et al., 2018). Only in *Pomacea* have PV1 proteins been isolated and characterized.

pigments, are high molecular weight oligomers, are highly glycosylated, and are composed of combinations of subunits with similar amino acid sequences (Dreon et al., 2004a; Ituarte et al., 2008; Pasquevich et al., 2014; Brola et al., 2020). Members of this novel family of invertebrate egg reproductive proteins also display high thermal and pH structural stability, as well as a kinetic stability (Pasquevich et al., 2017; Brola et al., 2020).

PV1s are massively accumulated in eggs (Giglio et al., 2016), playing a role as a storage protein and a major source of nutrients during embryo development (Heras et al., 1998). Moreover, PV1s carry and protect antioxidant carotenoid pigments from the harsh environmental conditions of development (Dreon et al., 2004a; Ituarte et al., 2008; Pasquevich et al., 2014). Remarkably, they are a poor amino acid source to predators because of their low digestibility, which renders them an antinutritive protein (Pasquevich et al., 2017; Brola et al., 2020). Noteworthy, besides embryo nutrition and the antinutritive (non-digestive) role, PV1s have other clade-related functions according to their phylogenetic position: those PV1s from the Canaliculata clade, PcOvo from *P. canaliculata* and PmPV1 from *P. maculata*, provide a bright reddish coloration, possibly a warning signal, an ecological function that would go along with the presence of a toxic perivitellin, Perivitellin-2 (PV2), only found in the Canaliculata clade (Giglio et al., 2020; Heras et al., 2008).

In contrast, the members of the Bridgesii clade lay more pale eggs of a pinkish color (presumably a non-warning signal) and have PV1s like PsSC from *P. scalaris* (Ituarte et al., 2008, 2010, 2012) and PdPV1 from *P. diffusa* (Brola et al., 2020) that possess a strong lectin activity, i.e. capacity to recognize and bind glycans, and adversely affect gut morphophysiology of predators, a role absent in PV1s from the Canaliculata clade. These different functions among orthologous PV1 proteins (Brola et al., 2020) go along with their remarkably high stability and provide a unique and unexplored model to understand the evolution of hyperstable proteins, i.e. proteins resistant to degradation, even under relatively harsh conditions (Colón et al., 2017). The evolution of hyperstable proteins has been poorly studied experimentally and the remarkable stability and their varied functions in the most derived clades make *Pomacea* egg PV1s excellent candidates for exploring this issue.

This prompted us to study whether the different structures and defensive roles of perivitellins were associated with their phylogeny. We began by examining the structure, stability and functional features of PV1s from Flagellata, the most basal clade of the genus, and determined some structural features from PsSC, PmPV1 and PcOvo. This approach allowed for a phylogenetic analysis of the clade-related structural and functional trends in these hyperstable proteins, providing one of the few available examples taken from nature. Our findings reveal a loss of structural and kinetic stability along ortholog evolution associated with the acquisition of new defensive traits. We found that variations in these reproductive proteins accompanied the diversification and radiation of the genus, potentially contributing to the emergence of some apple snails as notorious invasive pests.

MATERIALS AND METHODS

Sample collection, species identification and PV1 purification

Pomacea patula catemacensis eggs were collected in the Catemaco Lake, Veracruz, Mexico (18°24′14.75″N, 95°04′52.80″W) and kept in the laboratory at −70°C until processed. Genetic identification was performed as reported for other *Pomacea* species (Pasquevich and Heras, 2020). Briefly, DNA was extracted from embryos, using the Quick Tissue/Culture cells genomic DNA extraction kit (DSBIO) following the manufacturer's recommendations and then quantified by Nanodrop. The *MT-COI* gene was amplified by PCR using the primers proposed by Folmer et al. (1994), extensively employed for *Pomacea MT-COI* amplification (Hayes et al., 2009; Yang et al., 2018). PCR products were purified and sequenced by Macrogen (Seoul, South Korea). Sequences were then subjected to BLAST analysis (Altschul et al., 1997) to obtain species identification by comparison with sequences from individuals previously identified.

The perivitelline fluid (PVF) was obtained as previously described (Pasquevich et al., 2014). In short, two pools of three egg masses each were homogenized separately on ice in 20 mmol l^{-1} Tris-HCl buffer 1:3 (w/v) and sequentially centrifuged at 4°C for 20 min at 10,000 ***g*** and 50 min at 100,000 ***g***. We estimated each clutch to contain at least 300 eggs (Hayes et al., 2015). The obtained supernatant contained the soluble egg fraction.

To compare and trace the evolution of PV1 carotenoproteins from *Pomacea*, two pools of PV1 from *Pomacea patula catemacensis* (hereafter PpaPV1) were purified following the procedure described for other apple snail egg carotenoproteins (see below). This approach

ensured the incorporation of natural protein heterogeneity, encompassing post-transcriptional modifications and variability in oligomer assembly. Total protein was quantified following the method described by Bradford using bovine serum albumin (BSA, Sigma cat. no. 7906) as standard. Purity was checked by polyacrylamide gel electrophoresis (PAGE). Other *Pomacea* spp. carotenoproteins (i.e. PcOvo, PmPV1 and PsSC), used in some assays, were similarly purified (Dreon et al., 2004b; Ituarte et al., 2008; Pasquevich et al., 2014).

Structure

Oligomer and subunit electrophoretic behavior

Gel electrophoresis was used to characterize and compare the oligomer and subunits of PpaPV1 with other *Pomacea* PV1s. Native (non-denaturing) PAGE with Laemmli buffer (pH 8.8) without SDS was performed in 4–20% gradient polyacrylamide gels in a miniVE Electrophoresis System (GE Healthcare Life Science). High molecular weight standards (Pharmacia) were run in the same gels. Subunits were separated by SDS-PAGE in 4–20% gradient polyacrylamide gels containing 0.1% SDS; samples were denatured at 95°C, with dithiothreitol and β-mercaptoethanol treatment (Laemmli, 1970). Low molecular weight standards (Pharmacia) were used, and gels were stained with Coomassie Brilliant Blue R-250 (Sigma Chemicals). In both gels, PcOvo, PmPV1 and PsSC were run for comparison of PV1s from other clades.

Immunoblotting

Antibodies directed against purified PpaPV1 were prepared in rabbits at the Facultad de Agronomía of Universidad Nacional de La Plata. Animals were given a first subcutaneous injection of 120 µg of PpaPv1 emulsified in Freund's complete adjuvant (Sigma Chemicals, St Louis, MO, USA). A booster injection with about 60 µg antigen mixed with Freund's incomplete adjuvant was administered after 2 and 4 weeks. Two weeks later, rabbits were bled through cardiac puncture. The collected blood was allowed to clot overnight (4°C) and after centrifugation the serum obtained was stored at −70°C and used in the western blot technique. The specificity of the rabbit antiserum against PpaPV1 was verified by immunoblotting PpaPV1 with a non-immunized rabbit serum (Fig. S1A,B).

Antibody cross-reactivity for PpaPV1 with PV1s from other clades was assayed with anti-PpaPV1 prepared for this study (see paragraph above), anti-PsSC (Ituarte et al., 2008) and anti-PcOvo sera (Dreon et al., 2003) (Fig. S1C,D). PV1s (7 µg) and molecular weight marker (Dual Color, Bio-Rad) were separated by SDS-PAGE in 16% gels and transferred onto nitrocellulose membranes (Amersham) in a Mini Transblot Cell (Bio-Rad), using 25 mmol l^{-1} Tris-HCl, 192 mmol l^{-1} glycine, 20% (v/v) methanol, pH 8.3 buffer. After blocking for 2 h at room temperature with 3% (w/v) non-fat dried milk in PBS-Tween, the membranes were incubated overnight at 4°C with polyclonal antibodies against PcOvo (1:10,000 dilution), PsSC (1:12,000 dilution) and PpaPv1 (1:1000 dilution) in 3% (w/v) non-fat dried milk in PBS-Tween. Specific antigens were detected after incubation for 2 h at room temperature with 1:3000 dilution of goat anti-rabbit IgG horseradish peroxidase conjugate (Bio-Rad, cat. no. 172-1019) in 3% (w/v) non-fat dried milk in PBS-Tween. Immunoreactivity was visualized by electro-chemiluminescence in a Chemi-Doc MP Imaging System (Bio-Rad).

Size exclusion chromatography

Size exclusion chromatography (SEC) allows us to estimate the molecular weight of proteins by comparing the chromatographic retention times with those of standard proteins of precise weight (Barth and Boyes, 1990). The molecular weight (MW) of PV1s by SEC was analyzed with an SEC-Superdex 200 10/300 GL column (Amersham) in an isocratic size exclusion HPLC (1260 Infinity, Agilent Technologies) with UV detection in isocratic mode. The mobile phase contained 137 mmol l^{-1} NaCl, 2.7 mmol l^{-1} KCl, 10 mmol l^{-1} Na_2HPO_4 and 1.8 mmol l^{-1} KH_2PO_4 (in PBS). The flow rate was 0.5 ml min^{-1} and the detector was set at 280 nm. The elution volume (V_e) of PV1s (PpaPV1, PsSC, PmPV1 and PcOvo) and standard proteins in PBS [5 mg ml^{-1} thyroglobulin (MW 669,000), 2.8 mg ml^{-1} ferritin (MW 440,000), 4 mg ml^{-1} aldolase (MW 159,000) and ovoalbumin (MW 45,000)] was determined by measuring the volume of the eluent from the point of injection to the center of the elution peak. Blue Dextran 2000 (1 mg ml^{-1}) was used to calculate the column void volume (V_0). The average distribution constant, K_{av}, was used to normalize the elution behavior. A calibration curve was obtained by fitting a curve in a plot of $K_{av}=(V_e-V_0)/(V_c-V_0)$, where V_c is the geometric volume of the column (24 ml), versus the log molecular weight of each standard. A standard curve was fitted to a line and the MW of PV1s was calculated by extrapolating from the standard curve (GraphPad Prism version 8.0.1 for Windows, GraphPad Software, San Diego, CA, USA; www.graphpad.com).

Dynamic light scattering

The dynamic light scattering (DLS) of a nanoparticle sample in solution reveals the particle size distribution in real time (Falke and Betzel, 2019). DLS is particularly sensitive to large aggregates, common in some PV1s (Ituarte et al., 2008; Brola et al., 2020). Thus, DLS analysis was performed as soon as PV1s were eluted from SEC columns. PV1s sizes were monitored in PBS at 25°C, using a Malvern Zetasizer nano-zs instrument. The scatter light signals were collected at a 173 deg scattering angle (backscatter) and three measurements of an automatic number of runs each were conducted per sample. Protein parameters were analyzed with Zetasizer Software v.7.13. Data used for size measurement met quality criteria. Intensity size distributions were used for size calculation. Volume size distribution was used to check the main peak contribution to the analysis.

N-terminal sequence

Subunits of purified PpaPV1 separated by electrophoresis as above were transferred to PVDF membranes and sequenced by Edman degradation at the Laboratorio Nacional de Investigación y Servicios en Péptidos y Proteínas (LANAIS-PRO, Universidad de Buenos Aires – CONICET). The system used was an Applied Biosystems 477a Protein/Peptide Sequencer interfaced with an HPLC 120 for one-line phenylthiohydantoin amino acid analysis. N-terminal sequences were compared with other *Pomacea* sequences using the multiple sequence alignment program CLUSTAL 2.1 (Larkin et al., 2007).

Spectrophotometric analysis

Absorption spectra of egg carotenoproteins are valuable taxonomic characters in *Pomacea* spp. (Pasquevich and Heras, 2020). PV1s absorb light in the visible region of the spectrum as a result of their carotenoid pigments (Heras et al., 2007). Absorption spectra of PVF and purified carotenoproteins were recorded between 350 and 650 nm in an Agilent 8453 UV/Vis diode array spectrophotometer (Agilent Technologies, Waldbronn, Germany).

Structural and kinetic stability

Effect of pH and temperature on structural stability

To study the effect of pH on PpaPV1 structural stability, the protein was incubated overnight in different buffers ranging from pH 2.0 to

12.0 following a previously used method (Pasquevich et al., 2017). Samples were analyzed by absorbance and fluorescence spectroscopy. Absorbance spectra were recorded between 300 and 650 nm in an Agilent 8453 UV/Vis diode array spectrophotometer (Agilent Technologies, Waldbronn, Germany) taking advantage of the fact that PV1s absorb in the visible range, allowing us to follow the protein–carotenoid interaction by its spectrum in this range. Fluorescence emission was recorded as described in 'Chemical denaturation' (see below). Two independent samples were measured, and the corresponding buffer blank was subtracted. The effect of temperature on PpaPV1 at pH 7.4 was also measured by absorption and fluorescence spectroscopy in the range 25–85°C. The effect of extreme thermal conditions was analyzed by boiling PpaPV1 for 50 min and evaluating oligomer integrity using native (non-denaturing) PAGE, as previously done (Pasquevich et al., 2017).

Chemical denaturation

The intrinsic fluorescence emission of PpaPV1 and PsSC tryptophans was used to follow PV1 denaturation induced by guanidine hydrochloride (GndHCl) (Sigma). Chemical denaturation was performed by incubating PV1s (50 µg ml^{-1}) overnight in the presence of increasing concentrations (0–6.5 mol l^{-1}) of GndHCl buffered with 0.1 mol l^{-1} phosphate buffer, pH 7.4 at 8°C.

Protein intrinsic fluorescence spectra were recorded on a Fluorolog 3 Spectrofluorometer coupled with a Lauda Alpha RA 8 thermostatic bath. Fluorescence spectra were recorded in emission scanning mode at 25°C. Tryptophan emission was excited at 295 nm (6 nm slit) and recorded between 310 and 450 nm (3 nm slit). The corresponding buffer blank was subtracted. Two independent samples were measured. Spectra were characterized by their center of mass (CM) and the populations associated with the unfolded fraction (f_u) were calculated from the CM as for PmPV1 in Pasquevich et al. (2017).

The equilibrium reached in each GndHCl concentration allowed the calculation of an equilibrium constant $K=f_u/(1-f_u)$ and Gibb's free energy for the unfolding reaction in terms of this mole fraction ($\Delta G^0=-RT\ln K$, where R is the gas constant and T is temperature) were calculated. The dependence of ΔG^0 on GndHCl concentration can be approximated by the linear equation $\Delta G^0=\Delta G^0H_2O-m$[GndHCl], where the free energy of unfolding in the absence of denaturant (ΔG^0H_2O) represents the conformational stability of the protein. m is a parameter that correlates with the change in the solvent accessible area. The GndHCl concentration in which half of the protein is unfolded (C_m) was estimated as a function of denaturant concentration from the linear extrapolation method.

Resistance to SDS

Resistance to SDS-induced denaturation serves to identify proteins whose native conformations are kinetically trapped in a specific conformation because of an unusually high unfolding barrier that results in very slow unfolding rates. The resistance to SDS was assayed following the Manning and Colón procedure previously used with other *Pomacea* spp. carotenoproteins (Pasquevich et al., 2017; Brola et al., 2020). Briefly, PcOvo, PmPV1, PsSC and PpaPV1 were incubated in Laemmli sample buffer (pH 6.8) containing 1% SDS and either boiled for 10 min or unheated before analysis by 4–20% SDS-PAGE. The gels were then stained with Coomassie Brilliant Blue R-250.

Unfolding kinetics induced by GndHCl

Unfolding of proteins in increasing concentrations of GndHCl allows us to calculate the rate of unfolding (half-life) in the absence of denaturant under native conditions (Manning and Colón, 2004). A fluorolog-3 (Horiba–Jobin Yvon) fluorometer was used to measure the kinetics of PpaPV1, PsSC and PmPV1 unfolding. For the fluorescence kinetic experiment, protein solutions in 100 mmol l^{-1} phosphate buffer pH 7.4 (PB) were treated with increasing concentrations of GndHCl solution in the same buffer in a 10 mm pathlength cuvette. PV1s (final concentration of 50 µg ml^{-1}) were mixed with GndHCl by pipetting up and down with a 1 ml pipette. Data collection was started after the chamber was closed. The shutters open automatically. Time zero was manually determined as 10 s after the protein was added to the denaturant. The excitation/emission wavelengths were 299/360 nm. The relaxation time was fitted to an exponential equation (GraphPad Prism version 8.0.1 for Windows). The unfolding constants were obtained for each GndHCl concentration. The rate constants as a function of GndHCl were extrapolated to native conditions to obtain an estimate of the rate constant (k) in the absence of a denaturant. Half-life was calculated as ln2/k.

Resistance to proteolysis: proteinase K assay

Protein structural rigidity makes proteins resistant to proteolysis. The rigidity of PV1s was assayed by proteinase K treatment, performed following Kim et al. (2004) using the concentrations modified by Frassa et al. (2010) and previously performed on PmPV1 (Pasquevich et al., 2017). PpaPV1 (1 mg ml^{-1}) was incubated with proteinase K (1, 10 and 100 µg ml^{-1}) in 50 mmol l^{-1} Tris-HCl buffer (pH 8.0) containing 10 mmol l^{-1} $CaCl_2$ at 37°C for 30 min. Digestion was ended by boiling samples in SDS sample buffer, and products were analyzed by SDS-PAGE as above.

Functions of PpaPV1

Hemagglutinating activity

PV1s of the Bridgesii clade have a strong lectin activity and ability to agglutinate rabbit erythrocytes (Ituarte et al., 2012; Brola et al., 2020), but PV1s of the Canaliculata clade lack this capacity (Pasquevich et al., 2017). We tested this capacity in *P. patula* PpaPV1 using the same methodology. In short, PpaPV1 hemagglutinating activity was tested by hemagglutination of red blood cells (RBCs) from rabbits obtained in facilities of the University of La Plata. Erythrocytes were prepared as stated in Ituarte et al. (2012) with minimal modifications. Twofold serial dilutions of PpaPV1 in PBS (25 µl) were incubated with an equal volume of 2% (v/v) erythrocytes in PBS in U-shaped microtiter plates (Greiner Bio-One) at 37°C for 2 h. The results are expressed as the lowest protein concentration showing visible hemagglutinating activity by the naked eye.

Specificity for glycan binding

Glycan binding specificity of PpaPV1 was determined at the Core H of the Consortium for Functional Glycomics, CFG (http://www.functionalglycomics.org; Emory University, Atlanta, GA, USA). To detect the primary binding of PpaPV1 to glycans, proteins were fluorescently labeled using the Alexa Fluor 488 Protein Labeling kit (Invitrogen Molecular Probes, Life Technologies) according to the manufacturer's instructions and according to the CFG guidelines. Protein concentration and the degree of labeling were determined spectrophotometrically. Fluorescently labeled PpaPV1 was assayed on a glycan array that comprised 585 glycan targets (v.5.4). The results of glycan specificity for PpaPV1 were compared with those of PsSC (Ituarte et al., 2018) and PdPV1 (Brola et al., 2020) using the Glycan Array Dashboard (glycotoolkit.com/GLAD/) (Mehta and Cummings, 2019).

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In vitro intestinal digestion and high proteolysis assays

The simulated gastroduodenal digestion of PpaPV1 was analyzed by sequentially incubating the protein for 2 h with pepsin (gastric) and 2 h with trypsin (intestinal) at 37°C, using the method described by Moreno et al. (2005), with some modifications as described in Pasquevich et al. (2017). Briefly, for *in vitro* gastric digestion, PpaPV1 in double-distilled water was dissolved in simulated gastric fluid (SGF; 0.15 mol l^{-1} NaCl, pH 2.5) to a final concentration of 0.5 µg $µl^{-1}$. Digestion commenced by adding porcine pepsin (Sigma, cat. no. P6887) at an enzyme:substrate ratio of 1:20 (w/w). Gastric digestion was conducted at 37°C with shaking for 120 min. Aliquots of 5 µg protein were taken at 0, 60 and 120 min for SDS-PAGE. The reaction was stopped by increasing the pH with 150 mmol l^{-1} Tris-HCl buffer pH 8.5. Samples were immediately boiled for 5 min in SDS electrophoresis buffer with β-mercaptoethanol (4%) and analyzed as described above.

For *in vitro* duodenal digestion, the gastric digest was used as starting material. The digest was adjusted to pH 8.5 (as above) and sodium taurocholate (Sigma) was added. The simulated duodenal digestion was conducted at 37°C with shaking using bovine pancreas trypsin (Sigma, cat. no. T9935) at an enzyme:substrate ratio of 1:2.8 (w/w). Aliquots were taken at 0, 60 and 120 min for SDS-PAGE analysis. BSA was used as positive (with enzyme) and negative (without enzyme) control in both gastric and duodenal digestion.

Statistical analysis

Data were analyzed by first testing for normality using the Shapiro–Wilk normality test (α=0.05), and one-way ANOVA with Welch's correction. For glycan array data, one-way ANOVA with Welch's correction was followed by Games–Howell's multiple comparison test. For analysis, GraphPad Prism v.8.0.1 was used.

RESULTS

Genetic identification

Species identity was genetically assessed by obtaining the partial mitochondrially encoded-cytochrome *c* oxidase subunit I (MT-COI) sequence from eggs. A 670 bp PCR product was obtained, and a BLAST analysis aligned it with the GenBank sequence FJ710326.1, corresponding to '*Pomacea patula* isolate PA03 cytochrome *c* oxidase subunit I (COI) gene, partial cds; mitochondrial'. Although *P. patula* is the only *Pomacea* species reported in Catemaco Lake, the genetic analysis provided further confirmation of the species identity. Hereafter, *Pomacea patula* will refer to *P. patula catemacensis* for simplicity.

Structural features

Native PpaPV1 is an oligomer with a MW of ∼271 kDa, not different from that of other PV1s (Table 1). Similarly, when we determined the size of PV1s from the three clades using DLS, they showed a single main peak in the distribution of intensity and volume, ranging from 12.2 to 14.0 nm in diameter that was also not different among PV1s (Fig. 2A, Table 1; Fig. S2). The minor intensity peak at high diameter observed in PV1s from the Bridgesii and Canaliculata clades was ascribed to aggregation, which typically results from the formation of aggregated particles and is related to the intrinsic properties of the proteins (Fig. S2B); this aggregation is common in PV1s. Under native conditions, the electrophoretic mobility of PpaPV1 differed from that of PV1s of the derived clades (Fig. 2B). Upon reduction, PpaPV1 showed several subunits ranging between 25 and 35 kDa (Fig. 2B), the same MW range reported for the other PV1 of the genus.

Table 1. Molecular weight and size of apple snail PV1 perivitellins

PV1 protein	MW (kDa)	Size (diameter, nm)
PpaPV1	270.9±9.8	12.2±0.04
PsSC	272.0±7.5	14.0±0.46
PcOvo	278.0±16.6	13.1±0.41
PmPV1	268.8±9.5	12.4±0.37

Values were estimated by HPLC-size-exclusion chromatography (SEC; MW) and dynamic light scattering (DLS; size) and represent the mean±s.d. Regardless of the clade, apple snail PV1 proteins have no significant differences in MW or size (see Materials and Methods for details on statistical analysis).

Anti-sera against PpaPV1 (Flagellata clade) and PsSC (Bridgesii clade) cross-react with PsSC and PpaPV1, but not with PmPV1 and PcOvo (Canaliculata clade) (Fig. 2C; Fig. S1) while PcOvo anti-serum recognized Canaliculata and Bridgesii clade PV1s (i.e. PcOvo, PmPV1 and PsSC) and also PdPV1 as previously reported (Dreon et al., 2003; Pasquevich et al., 2017; Brola et al., 2020) but not with PpaPV1 (Fig. 2C).

Under dissociating conditions, PpaPV1 was separated into seven subunits numbered according to their electrophoretic mobility. Remarkably, some of their N-terminal sequences were identical and therefore were grouped into two nearly identical sequences (Fig. S3A) before multiple sequence analysis (Fig. S3B). Sequence similarity analysis showed the PpaPV1-A1 group had 64.7% similarity with PdPV1 and 70.6% similarity with the other PV1s (Fig. S3C), while the PpaPV1-A3 group of sequences presents 70.6% similarity with PsSC and 66.7 with PdPV1 (Bridgesii clade) and 55.6% with PmPV1 and PcOvo (Canaliculata clade) (Fig. S3D). Remarkably, PpaPV1 band 4 N-terminal sequences had no similarity with any reported *Pomacea* spp. PV1 Perivitellin.

Spectroscopic measurements provided further insight into the structure, indicating that both PpaPV1 and egg PVF absorb across a wide range of the visible spectra (350–650 nm) with a maximum at 380 nm. This absorption maximum is shared with the Bridgesii clade (*P. scalaris*) but differs from that of the Canaliculata clade PV1s (Fig. 3). Additionally, the overall relative absorption intensity between 450 and 600 nm increases in accordance with the phylogenetic position, from the Flagellata clade towards the Canaliculata clade (Fig. 3).

Structural and kinetic stability

Structural stability against pH, temperature and chemical chaotropes

PpaPV1 remained stable in a wide range of pH. A slight alteration in the fine structure of the UV-visible spectrum (Fig. S4A) and an increase in fluorescence emission intensity (Fig. S4B) were only observed at pH 2.0. The absorption and emission spectra of PpaPV1 remained virtually unchanged even at temperatures of 80–85°C (Fig. S4C,D). Also, the electrophoretic behavior after boiling PpaPV1 for 60 min was unchanged, as reported for other *Pomacea* carotenoproteins (Fig. S4E).

The chemical stability of PV1s showed an overall increase in fluorescence intensity and a systematic red shift of the spectra when increasing GndHCl concentration (Fig. S5). Fig. 4A shows that the GndHCl unfolding transition of PpaPV1 reached a plateau and the experimental data fitted a two-state model. The GndHCl concentration required to reach the midpoint of the transition between the two states (C_m) was lower in PmPV1 and PsSC than in PpaPV1 (the GndHCl concentration to obtain 50% unfolded) (Fig. 4B, Table 2). The disassembling/unfolding process followed by changes in the standard free energy ΔG^0H_2O was higher in PpaPV1 than in PmPV1 (Fig. 4B). Table 2 shows this parameter in different *Pomacea* spp. PV1s.

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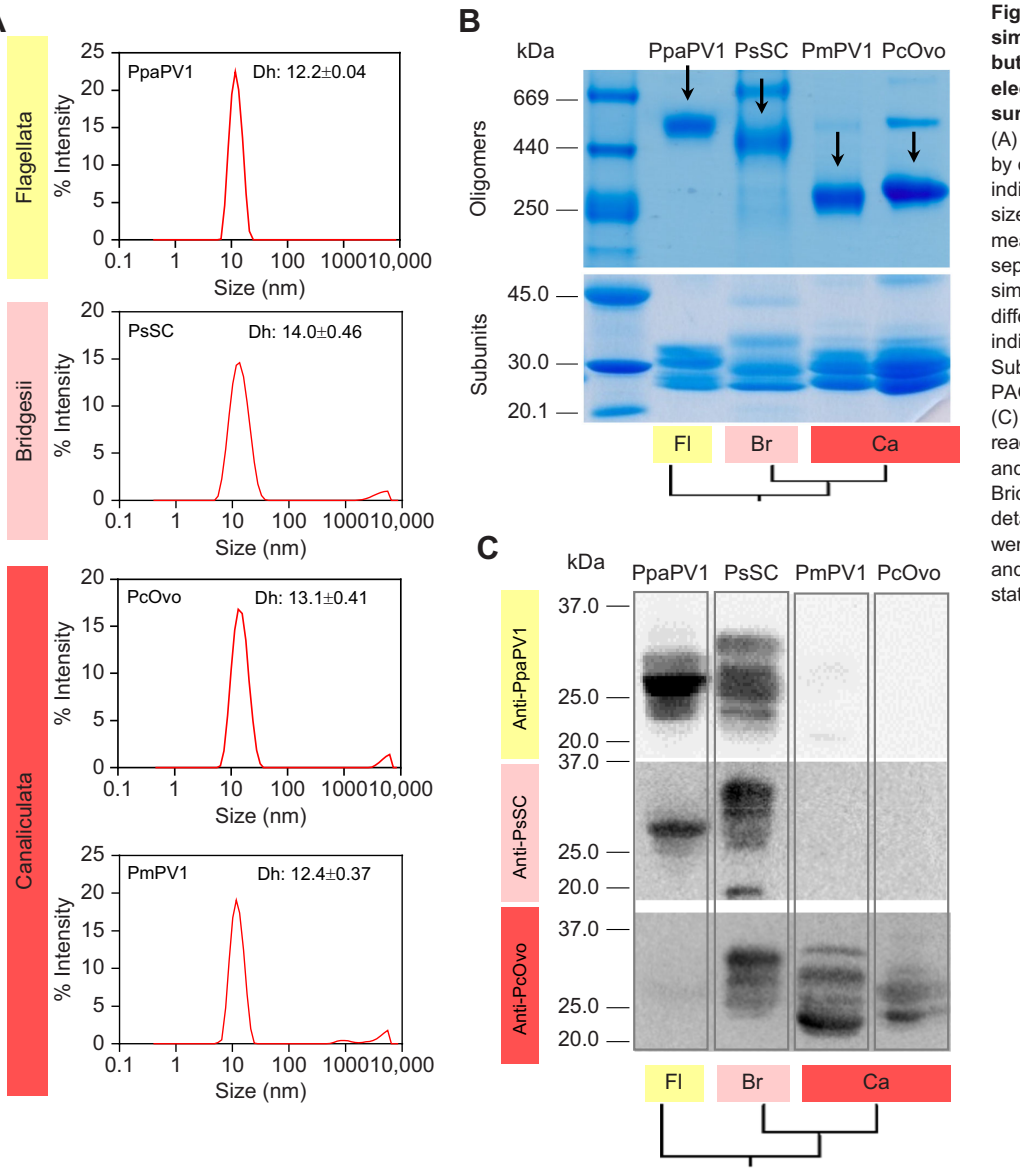


Fig. 2. Apple snail egg PV1s have a similar size and number of subunits, but show clade-related differences in electrophoretic migration, charge surface and immune cross-reactivity. (A) Particle size analysis as determined by dynamic light scattering (DLS) indicates that all PV1s have a similar size [hydrodynamic diameter (Dh); means±s.d.]. (B) Oligomers of PV1s separated by native PAGE (top) have similar molecular weight (MW) but different electrophoretic behavior (arrows indicate the relative mobility of PV1s). Subunits of PV1s separated by SDS-PAGE have a similar MW range (bottom). (C) Western blot analysis for cross-reactivity using anti-PpaPV1, anti-PsSC and anti-PcOvo sera. Fl, Flagellata; Br, Bridgesii; Ca, Canaliculata. See Fig. 1 for details on clades and species. Samples were pooled for B and C (see Materials and Methods for experimental details and statistical analysis).

Resistance to SDS-induced denaturation

Proteins with a high energetic barrier between the folded and unfolded states are very resistant to unfolding and are considered kinetically stable. The SDS-resistance assay was proposed as a method to determine kinetic stability. Comparison of the migration on polyacrylamide gels of PV1s incubated in SDS and then boiled or not boiled indicated PpaPV1 and, to some extent, PsSC were resistant to SDS-induced denaturation whereas PmPV1 and PcOVO partially disaggregated, displaying a partial loss and therefore some oligomers in the unheated samples disaggregated into their subunits (Fig. 4C).

Unfolding rate and resistance to proteolysis: kinetic stability of PV1s

Although all PV1 proteins are SDS resistant (Fig. 4C), when PV1s of the Flagellata, Bridgesii and Canaliculata clades were assayed, we observed that the rate at which PpaPV1 carotenoprotein unfolded was markedly lower than that of its derived clade counterparts, therefore being several orders more stable than PV1s of the Bridgesii (PsSC)

Table 2. Conformational stability of *Pomacea* apple snail perivitellins determined by the unfolding induced by chemical treatment with guanidine hydrochloride (GndHCl)

	Flagellata	Bridgesii		Canaliculata	
Clade	*P. patula*	*P. scalaris*	*P. diffusa*	*P. maculata*	*P. canaliculata*
PV1	**PpaPV1**	**PsSC**	PdPV1‡	**PmPV1**	PcOVO*
ΔG^0H_2O (kcal mol^{-1})	**6.40±0.32**	**1.43±0.44**	1.14	**3.32±0.19**	3.48±0.06
C_m (mol l^{-1})	**5.4**	**4.1**	3.4	**2.8**	ND

Data from this study are in bold. ΔG^0H_2O, free energy of unfolding in the absence of denaturant represents the conformational stability of the protein. C_m, GndHCl concentration in which half of the protein is unfolded. Values are means±s.d. (n=2). Apple snail PV1 protein stability differed among the three clades (see Materials and Methods for experimental details and statistical analysis). *Dreon et al. (2007), ‡ Brola et al. (2020); ND, not determined.

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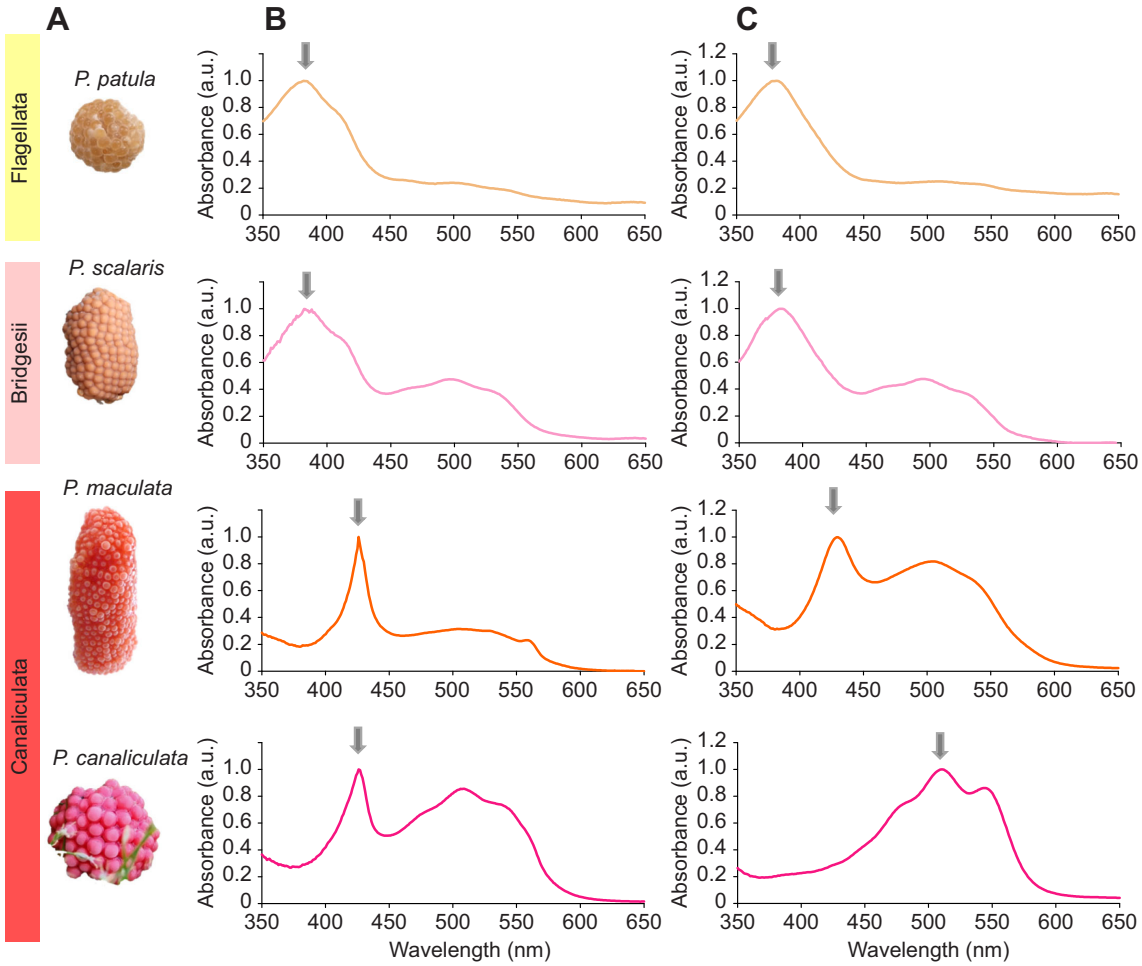


Fig. 3. *Pomacea* egg extract and PV1 absorption spectra shift toward red in most derived clades. (A) Egg masses. (B) Egg extract (perivitelline fluid, PVF). (C) Purified egg carotenoproteins. Spectra are ordered from basal (top) to derived (bottom) clades. Arrows indicate the maximum of each spectrum to highlight the red-shift from basal to derived clades. Spectra were normalized for ease of comparison. Data of *P. scalaris* egg carotenoprotein taken from Ituarte et al. (2008). Photo of *P. scalaris* eggs kindly provided by Ignacio Pavía.

and Canaliculata (PmPV1) clades (Fig. 4D). Consequently, the calculated half-life of PpaPV1 was several orders greater than those estimated for PsSC and PmPV1 (Fig. 4D).

Limited proteolysis of PpaPV1 with proteinase K, a fungal protease with broad specificity, showed no evidence of substantial PpaPV1 hydrolysis, while BSA (control) was completely digested. The low MW band that appeared after treating PpaPV1 with the highest concentration of proteinase K (100 µg ml^{-1}) is probably due to degraded proteinase K and/or products of PpaPV1 aggregation (Fig. S6A).

Functional characteristics

Lectin activity

The carbohydrate-binding capacity of PpaPV1 evaluated by hemagglutination assays showed activity in a dose-dependent manner. PpaPV1showed hemagglutinating activity to a concentration up to 0.26±0.148 µg µl^{-1} (Fig. 5A). The specificity and relative affinity of PpaPV1 towards oligosaccharide structures were evaluated by a high-throughput glycan array assay. PpaPV1showed a binding pattern (albeit with mild affinity) to glycans related to blood group A containing a specific motif [GalNAcα1-3(Fucα1-2)Galβ1-4GlcNAc] (Fig. 5B). The specificity toward oligosaccharides is shown in Table 3. Among them, the specificity of PpaPV1 with blood group A type 2 antigens, as well as the lack of specificity toward sialic acid antigens, are remarkable.

In vitro simulated gastrointestinal digestion

PpaPV1 resists hydrolysis when exposed sequentially to 2 h of gastric and duodenal phases (Fig. S6B) while BSA (control) was readily degraded. PpaPV1 maintained its electrophoretic behavior for up to 120 min.

DISCUSSION

The analysis of a PV1 protein in the most basal clade of the *Pomacea* genus (Hayes et al., 2009) has enabled, for the first time, an exploration of the evolutionary trajectories of structure, stability and functional features of ortholog proteins within a single genus. Fig. 6 summarizes the potential evolutionary trajectories of these PV1 traits across the *Pomacea* genus, discussed further below. One of the first conclusions is that PV1s purified from their natural source are highly stable and exhibit a loss of stability as they evolve, acquiring functions that co-evolve with characters known to favor the species' success (Hayes et al., 2015).

PV1s show a conserved general structure (mass and size) coupled with marked structural changes such as an increased subunit sequence diversity, post-translational modifications and surface

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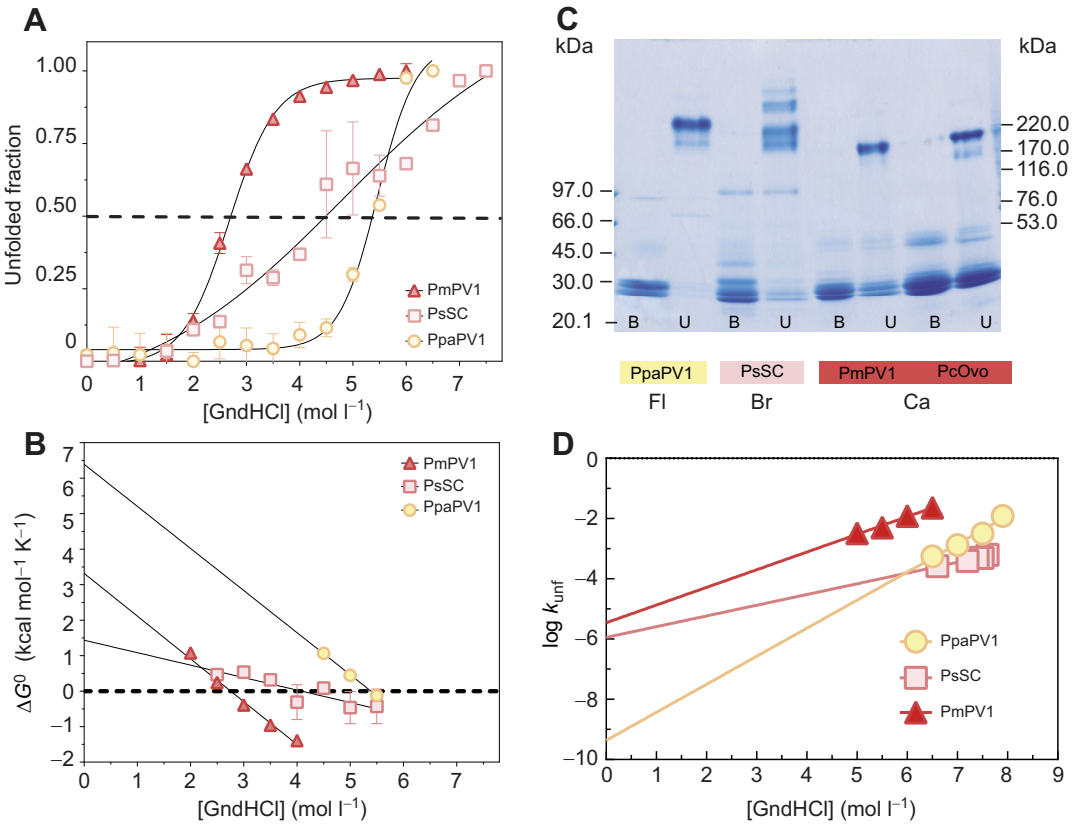


Fig. 4. Structural and kinetic stability of PV1 decrease in a clade-related fashion. Stability was evaluated by the unfolding induced by guanidine hydrochloride (GndHCl). (A) Unfolded population of PV1s at equilibrium. (B) Dependence of the unfolding free energy (ΔG^0) on GndHCl concentration. ΔG^0H_2O was calculated from the ordinate intercept. The dashed line indicates C_m, the GndHCl concentration at ΔG^0=0 (midpoint of the denaturing transition). PmPV1 data taken from Pasquevich et al. (2017). Data in A and B are means±s.d. (C) Representative SDS-PAGE of PV1s unheated (U) or boiled (B) in the presence of SDS detergent for 10 min and immediately loaded onto the gel. PpaPV1 is more resistant to detergent treatment than PV1s from more derived clades. Fl, Flagellata; Br, Bridgesii; Ca, Canaliculata. (D) Unfolding rates of PpaPV1, PsSC and PmPV1 under native-like conditions are shown by extrapolating the unfolding rate determined at different concentrations of GndHCl to 0 mol l^{-1}. Extrapolation of the plot to 0 mol l^{-1} GndHCl yielded a native unfolding rate (k) of 4.3×10^{-10} for PpaPV1 (i.e. ∼51 years), 1.14×10^{-6} for PsSC (i.e. ∼7 days) and 3.5×10^{-6} for PmPV1 (i.e. ∼2 days). PpaPV1 has unfolding kinetics much slower than the ortholog of the most derived clades. Samples were pooled for C and D (see Materials and Methods for experimental details and statistical analysis).

structural alterations that were phylogenetically acquired. This reflects an intricate tradeoff between the evolution of new functions and protein stability. Regarding subunit diversity, it has been reported that PV1 heteroligomers are composed of combinations of related subunits, probably paralogs that arise by duplication and speciation from an ancient orphan gene (Sun et al., 2012; Pasquevich et al., 2017; Ip et al., 2019; Brola et al., 2020). PpaPV1 has only three distinct N-terminal sequences, suggesting several post-translational modifications rendering the seven subunits observed by electrophoresis, though mutations in

Table 3. Main glycan structures recognized by PpaPV1 and their relative binding strength

Rank	Oligosaccharide structure	Average RFU	% CV
1	GalNAcα1-3(Fucα1-2)Galβ1-4GlcNAcβ1-2Manα1-6(GalNAcα1-3(Fucα1-2)Galβ1-4GlcNAcβ1-2Manα1-3)Manβ1-4GlcNAcβ1-4GlcNAcβ-Sp20	1081±73[a]	7
2	GalNAcα1-3(Fucα1-2)Galβ1-4GlcNAcβ1-3Galβ1-4GlcNAcβ-Sp0	325±40[b]	12
3	Galβ1-4GalNAcα1-3(Fucα1-2)Galβ1-4GlcNAcβ-Sp8	318±41[b]	13
4*	Galβ1-3GlcNAcβ1-3Galβ1-3GlcNAcβ-Sp0	309±3[b]	1
5	Galβ1-3GalNAcα1-3(Fucα1-2)Galβ1-4GlcNAc-Sp0	239±28[b]	12
6	GalNAcα1-3(Fucα1-2)Galβ1-4GlcNAcβ1-2Mana-Sp0	204±20[c]	10
7*	(3S)Galβ1-4(Fucα1-3)(6S)Glc-Sp0	163±14[c]	9

The common recognition patterns are underlined (shaded in yellow in Fig. 5). Binding intensities are expressed as the mean of relative fluorescence units (RFU)±1 s.d., n=4. %CV=100×s.d. Statistical analysis was performed using one-way ANOVA with Welch's correction, followed by Games–Howell's multiple comparison test. Different superscript letters indicate significant differences (P<0.05). Full data on PpaPV1 glycan specificity are available from the authors upon request. *Not PpaPV1 dose-dependent binding. The most common specific glycan motif [GalNAcα1-3(Fucα1-2)Galβ1-4GlcNAc] is underlined.

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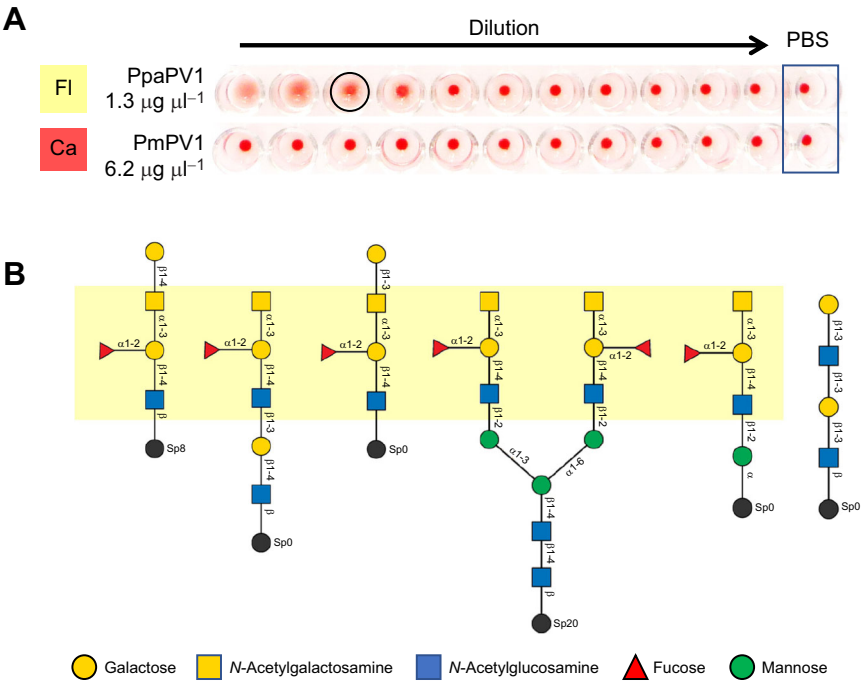


Fig. 5. Lectin activity and glycan-binding motifs of Flagellata PpaPV1. (A) Representative hemagglutinating activity of PpaPV1 from *P. patula*. Twofold serial dilutions of PV1s were incubated with red blood cells. The initial concentration of PV1s is listed on the left. The PBS control is on the right. The circled well corresponds to the lowest PpaPV1 concentration (i.e. highest dilution) showing visible hemagglutinating activity. PpaPV1 from Flagelatta (Fl) has moderate lectin activity. PV1 belonging to the derived Canaliculata (Ca) clade (PmPV1 from *P. maculata*) lacks this capacity. (B) The main glycan structures recognized by PpaPV1. Yellow shading highlights the common recognition pattern GalNAca1-3(Fuca1-2)Galb1-4GlcNAc. Glycan structures were taken from the Consortium for Functional Glycomics (https://www.functionalglycomics.org/). See Table 3 for details on PpaPV1 glycan specificity.

selected regions cannot be disregarded. Thus, the number of unique paralogs forming the oligomers doubled in derived clades (Pasquevich et al., 2017; Ip et al., 2019; Brola et al., 2020), suggesting strong selective pressure on structure and stability, leading to new or improved functions.

Furthermore, the spectroscopic features (and egg coloration) transition from a pale coloration in the most basal species to a bright pink–red coloration in the Canaliculata clade (Fig. 3A). This change in color (carotenoids) is associated with the development of new egg defenses (see below).

PpaPV1 exhibits unparalleled structural and kinetic stability, evidenced in the wide range of pH and temperatures it withstands, unfolding half-life, resistance to detergent unfolding, and retention of a rigid folded core that only unfolds when boiled for several minutes (Manning and Colón, 2004). Its kinetic stability is higher than that of PV1s from derived clades (Pasquevich et al., 2017; Brola et al., 2020). Kinetic stability in proteins plays a crucial role in numerous known and yet to be discovered biological functions via its ability to confer degradation resistance (Colón et al., 2017; Manning and Colón, 2004; Xia et al., 2007). In this regard, PV1s are remarkably resistant to SDS and proteolytic cleavage. All the other studied PV1s are also highly stable (Brola et al., 2020; Pasquevich et al., 2017). However, unexpectedly we found that PpaPV1 ranks among the most hyperstable proteins reported, according to the protein stability list compiled by Manning and Colón (2004). Along PV1 evolution, there was a partial loss of stability in PV1s of the more derived clades (Fig. 6). The high kinetic stability observed in this novel protein family appears to be a vital property for their protective role during embryo development, contributing to the reproductive success of apple snails in nature (Manning and Colón,

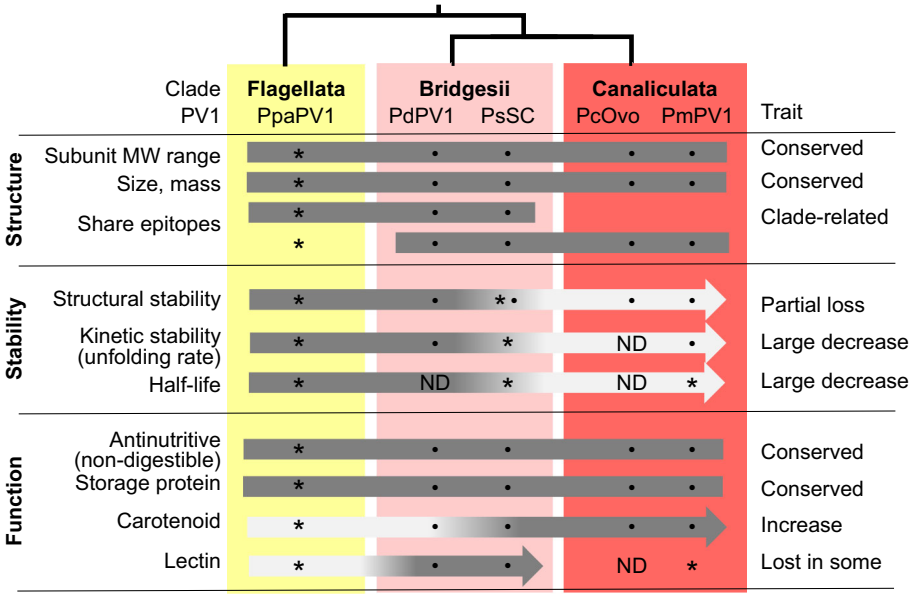


Fig. 6. Proposed evolutionary trajectory of structure, stability and functional features of PV1 carotenoproteins across the *Pomacea* genus. Dark gray bars indicate conserved characters (synapomorphy). Arrows indicate characters changing along clades, with darker shading representing increased trait. ND, not determined. *Data from this study. Circles indicate data taken from Brola et al. (2020) and Ituarte et al. (2008, 2010, 2012) for Bridgesii PV1s and Dreon et al. (2003, 2004a,b, 2010) and Pasquevich et al. (2014, 2017) for Canaliculata PV1s. Please refer to Fig. 1 for details on species and clades.

2004). Considering reports suggesting that high stability facilitates the evolvability of proteins (Bloom et al., 2006), it can be argued that the hyperstability of PpaPV1 may have provided tolerance to mutations, allowing the acquisition of new functions (functional evolution) during *Pomacea* diversification without compromising the native structure. The last section of this discussion will delve into the diverse functions that PV1 proteins have evolved.

Terrestrial egg deposition in the amphibious *Pomacea* was a key adaptation to avoiding aquatic predation and/or parasitism (Sun et al., 2019). This evolutionary driver influenced the defenses of snail eggs, and under this selective pressure, PV1 orthologs underwent significant functional adaptations while retaining some ancestral roles Thus, PV1 glyco-carotenoproteins kept their ancestral traits as storage proteins for nurturing embryos (probably by co-evolution with specific glycosidases) and retained their indigestibility by predators. However, evolutionarily these proteins experienced a partial loss of stability as they gained new functions (Fig. 6), illustrating a well-known tradeoff between the evolution of new functions and protein stability (Tokuriki and Tawfik, 2009b; Jayaraman et al., 2022). This evolutionary pathway was probably influenced by other proteins and biomolecules with which they interact and co-evolve (Jayaraman et al., 2022). The lower unfolding rate of PpaPV1 in the basal clade, compared with its orthologs in the most derived Canaliculata clade, probably facilitated structural changes. Some of these changes, such as the capacity of PcOvo to incorporate and withstand removal of large amounts of carotenoids without affecting its stability, are notable (Dreon et al., 2007).

This natural evolution aligns with laboratory and simulated evolution studies suggesting that the loss of stability could contribute to the acquisition of new protein functions (Bloom et al., 2006). However, this PV1 feature came at the expense of losing lectin capacity in the Canaliculata clade, which may initially seem puzzling given lectin's significant roles in defending eggs against predation (Brola et al., 2020) or plant seeds against herbivory (Peumans and Van Damme, 1995) (Fig. 6). However, this loss could be explained by the evolutionary novelty in the Canaliculata clade of a dual enterotoxic/neurotoxin lectin (PV2) combining two ancient immune proteins (Giglio et al., 2020; Sun et al., 2019). Conversely, the PV1s of the sister Bridgesii clade not only retained the lectin capacity of the PpaPV1 from the Flagellata clade but also evolved a higher affinity to glycans and a broader specificity indicative of at least two high-affinity carbohydrate recognition sites (Ituarte et al., 2018). Notably, Bridgesii PsSC and PdPV1 exhibit sugar recognition patterns that include Galβ1-3GaNAc and a common sialic acid in vertebrate gangliosides (Brola et al., 2020; Ituarte et al., 2018), absent in the glycan motifs recognized by PpaPV1. The inability of PpaPV1 to recognize gangliosides and its limited recognition patterns of group A type II antigens suggest the presence of only a single recognition site in this ancient PV1. It can be speculated that the partial loss of stability in PV1s of the Bridgesii clade may have favored the evolution of improved carbohydrate recognition domains and better binding strength capabilities. PV1 evolution is, to the best of our knowledge, one of the few examples taken from nature where the tradeoff between the stability and evolvability of a protein is reported.

Conclusion

Our work provides one of the first examples from natural evolution, rather than directed evolution in the laboratory, showing the crucial link between protein structure, stability and evolution of new functions.

We showcase that during ortholog evolution of a novel family of invertebrate reproductive proteins there is a tradeoff between a loss of structural and kinetic stability and the acquisition of new or improved defensive traits for snail embryos. The extent to which these mechanisms are evolutionary steps or alternative trajectories in the selective expression of defensive strategies in these snail eggs remains an open question.

Considering the relatively recent split of *Pomacea* from its sister genus about 29 mya (Sun et al., 2019), the study highlights how a rapid evolution of structure–function features of reproductive proteins accompanied the spread and diversification of *Pomacea* snails across freshwater habitats. Predator-induced protein evolution may have contributed to the evolved defense strategies, ultimately playing a role in the global invasiveness of Canaliculata snails.

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Competing interests

The authors declare no competing or financial interests.

Author contributions

Conceptualization: M.Y.P., M.S.D., M.E.D., H.H.; Methodology: M.Y.P., M.S.D.; Software: M.Y.P., H.H.; Validation: M.Y.P., M.S.D., H.H.; Formal analysis: M.Y.P., M.S.D., M.E.D., H.H.; Investigation: M.Y.P., M.S.D., M.E.D., H.H.; Resources: H.H.; Data curation: M.Y.P., H.H.; Writing - original draft: M.Y.P.; Writing - review & editing: M.Y.P., M.S.D., M.E.D., H.H.; Visualization: M.Y.P., M.S.D., H.H.; Supervision: M.Y.P., H.H.

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Data availability

All relevant data can be found within the article and its supplementary information.

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